

Quasi-exactly solvable systems and orthogonal polynomials

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This paper shows that there is a correspondence between quasi-exactly solvable models in quantum mechanics and sets of orthogonal polynomials $\{P_n\}$. The quantum-mechanical wave function is the generating function for the $P_n(E)$, which are polynomials in the energy E . The condition of quasi-exact solvability is reflected in the vanishing of the norm of all polynomials whose index n exceeds a critical value J . The zeros of the critical polynomial $P_J(E)$ are the quasi-exact energy eigenvalues of the system. © 1996 American Institute of Physics. [S0022-2488(96)03201-X]

In quantum mechanics there exist potentials for which it is possible to find a finite portion of the energy spectrum and associated eigenfunctions exactly and in closed form. These systems are said to be quasi-exactly solvable.¹⁻⁵ In such systems the potential depends on a parameter J ; for positive integer values of J one can find J eigenvalues and eigenfunctions exactly. The usual approach to the analysis of quasi-exactly solvable systems is an algebraic one in which the Hamiltonian is expressed as a nonlinear combination of generators of a Lie algebra, not belonging to the center of the corresponding enveloping algebra.⁴ This technique is a modification of the dynamical symmetry approach to exactly solvable quantum-mechanical systems, in which one can find by algebraic means the *entire* spectrum in closed form.⁶

In this paper we propose an alternative approach to quasi-exact solvability. We show that the solution ψ to the Schrödinger equation for a quasi-exactly solvable model,

$$H\psi = E\psi, \quad (1)$$

is the generating function for a set of polynomials $\{P_n(E)\}$ in the energy variable E . These polynomials satisfy a three-term recursion relation and therefore form an orthogonal set with respect to some weight function $w(E)$. For positive integer values of the parameter J , corresponding to quasi-exact solvability, we find that the norm of $P_n(E)$ vanishes for $n \geq J$. Moreover, all polynomials $P_n(E)$ beyond a critical polynomial $P_J(E)$, factor into a product of two polynomials, one of which is $P_J(E)$:

$$P_{n+J}(E) = P_J(E)Q_n(E) \quad (n \geq 0). \quad (2)$$

The zeros of the critical polynomial $P_J(E)$ are precisely the quasi-exact energy eigenvalues of the quantum-mechanical model.

We illustrate these features of quasi-exactly solvable models with the following infinite class of Hamiltonians first discussed by A. Turbiner:¹

$$H = -\frac{d^2}{dx^2} + \frac{(4s-1)(4s-3)}{4x^2} - (4s+4J-2)x^2 + x^6. \quad (3)$$

Here, s is an arbitrary parameter. When s lies between $\frac{1}{4}$ and $\frac{3}{4}$, there is an attractive centrifugal term; for s outside this range the centrifugal term is repulsive. When $s = \frac{1}{4}$ or $s = \frac{3}{4}$, the centrifugal core term disappears leaving a nonsingular sextic oscillator Hamiltonian

$$H = -\frac{d^2}{dx^2} - (4s + 4J - 2)x^2 + x^6. \quad (4)$$

When the parameter J in Eq. (3) is a nonnegative integer, the corresponding Schrödinger equation has J exact, closed-form solutions for any value of s .

We seek a solution $\psi(x)$ to the Schrödinger equation for H in Eq. (3) of the form

$$\psi(x) = \exp\left(-\frac{1}{4}x^4\right) x^{2s-1/2} \sum_{n=0}^{\infty} \left(-\frac{1}{4}\right)^n \frac{P_n(E)}{n! \Gamma(n+2s)} x^{2n}. \quad (5)$$

Observe that when $s = \frac{1}{4}$ this solution becomes an even-parity wave function of the oscillator Hamiltonian (4); when $s = \frac{3}{4}$, $\psi(x)$ becomes an odd-parity wave function of H in (4).

Demanding that $\psi(x)$ in Eq. (5) obey the Schrödinger equation (1) leads to the following recursion relation for the expansion coefficients $P_n(E)$:

$$P_n(E) = EP_{n-1}(E) + 16(n-1)(n-J-1)(n+2s-2)P_{n-2}(E) \quad (n \geq 2), \quad (6)$$

subject to the initial conditions

$$P_0(E) = 1 \quad \text{and} \quad P_1(E) = E. \quad (7)$$

From these initial conditions the recursion relation (6) generates a set of monic⁷ polynomials, the next four of which are

$$\begin{aligned} P_2(E) &= E^2 + (32 - 32J)s, \\ P_3(E) &= E^3 + [(160 - 96J)s - 32J + 64]E, \\ P_4(E) &= E^4 + [(448 - 192J)s - 128J + 352]E^2 + (3072J^2 - 12288J + 9216)s(s+1), \\ P_5(E) &= E^5 + [(960 - 320J)s - 320J + 1120]E^3 + [(15360J^2 - 81920J + 91136)s^2 \\ &\quad + (25600J^2 - 141312J + 164864)s + 6144J^2 - 36864J + 49152]E. \end{aligned} \quad (8)$$

These polynomials have a number of noteworthy properties. First, for all values of the parameters s and J they form an orthogonal set. This follows from the fact that they are generated by a second-order (three-term) recursion relation.⁸ The appearance of a three-term recursion relation is a consequence of the form of the potential in Eq. (3). For example, the corresponding recursion relation for an x^4 anharmonic oscillator potential, whose Hamiltonian is not quasi-exactly solvable, is a higher-order recurrence relation. The harmonic oscillator system leads to a two-term recursion relation; this system is exactly solvable rather than quasi-exactly solvable.

Second, from the expansion (5) we can see that the wave function $\psi(x, E)$ is the generating function for the polynomials $P_n(E)$.

The third and most significant property of the polynomials $P_n(E)$ is that, when the parameter J takes positive integer values, the polynomials exhibit the factorization property in Eq. (2). This factorization occurs because the third term in the recursion relation (6) vanishes when $n = J + 1$, so that all subsequent polynomials have the common factor $P_J(E)$. This factorization property holds for all values of the parameter s . Furthermore, this factorization leads to the result that the zeros

of the critical polynomial $P_J(E)$ are just the quasi-exact energy eigenvalues. This is true because the expansion in (5) truncates when E is a zero of $P_J(E)$; when this series truncates the wave function $\psi(x)$ is automatically normalizable.

To illustrate this factorization we list in factored form the first six polynomials $P_n(E)$ for the case $J=3$:

$$\begin{aligned} P_0(E) &= 1, \quad P_1(E) = E, \quad P_2(E) = E^2 - 64s, \\ P_3(E) &= E^3 - (128s + 32)E, \quad P_4(E) = [E^3 - (128s + 32)E]E, \\ P_5(E) &= [E^3 - (128s + 32)E](E^2 + 128s + 192). \end{aligned} \quad (9)$$

Observe that $P_3(E)$ is a common factor of $P_n(E)$ for $n \geq 3$. The zeros of $P_3(E)$ are

$$E = 0, \quad E = \pm \sqrt{128s + 32}, \quad (10)$$

which are the three exact energy eigenvalues for the quasi-exactly solvable Hamiltonian (3) when $J=3$. The corresponding exact eigenfunctions are obtained by evaluating $\psi(x)$ in Eq. (5) at these values of E :

$$\begin{aligned} \psi_0(x) &= \exp\left(-\frac{1}{4}x^4\right) \frac{x^{2s-1/2}}{\Gamma(2s)} \left(1 - \frac{x^4}{2s+1}\right), \\ \psi_+(x) &= \exp\left(-\frac{1}{4}x^4\right) \frac{x^{2s-1/2}}{\Gamma(2s)} \left(1 - \frac{\sqrt{128s+32}}{8s}x^2 + \frac{x^4}{2s}\right), \\ \psi_-(x) &= \exp\left(-\frac{1}{4}x^4\right) \frac{x^{2s-1/2}}{\Gamma(2s)} \left(1 + \frac{\sqrt{128s+32}}{8s}x^2 + \frac{x^4}{2s}\right). \end{aligned} \quad (11)$$

Note that the energy levels may be ordered by the number of nodes of the corresponding wave function.

A fourth property of the polynomials $P_n(E)$ concerns their norms. The norm (squared) γ_n of $P_n(E)$ is defined as an integral:

$$\gamma_n = \int dE w(E) [P_n(E)]^2. \quad (12)$$

It is possible to determine the norms of an orthogonal set of polynomials directly from the recursion relation; it is not necessary to know explicitly the weight function $w(E)$ with respect to which the polynomials are orthogonal.⁹ The procedure is simply to multiply the recursion relation (6) by $w(E)E^{n-2}$ and to integrate with respect to E . Using the fact that $P_n(E)$ is orthogonal to E^k , $k < n$, we obtain a simple, two-term recursion relation for γ_n :

$$\gamma_n = 16n(J-n)(2s+n-1)\gamma_{n-1}. \quad (13)$$

The solution to this equation with $\gamma_0 = 1$ is

$$\gamma_n = \frac{16^n n! \Gamma(J) \Gamma(2s+n)}{\Gamma(J-n) \Gamma(2s)}. \quad (14)$$

This equation reveals that the space of orthogonal polynomials arising from a quasi-exactly solvable model is associated with a nonpositive definite norm. In particular, we can see from Eq.

(14) that γ_n vanishes for $n \geq J$ if J is a positive integer. The appearance of a vanishing norm coincides with the factorization mentioned above and is an alternative characterization of quasi-exact solvability.

It is interesting that while the polynomials $P_{n+J}(E)$ for $n \geq 0$ have vanishing norm when J is a positive integer, the quotient polynomials $Q_n(E)$ in Eq. (2) form a new orthogonal set of polynomials for each value of J .

Having determined the norms γ_n of the polynomials $P_n(E)$ it is natural to evaluate the integral of the square of the generating function (wave function) with respect to the weight function $w(E)$:

$$G(x) = \int dE w(E) [\psi(x, E)]^2, \quad (15)$$

where $\psi(x, E)$ is given in Eq. (5). Using the orthogonality of the polynomials $P_n(E)$, we can express $G(x)$ as a confluent hypergeometric function:

$$G(x) = \frac{\Gamma(J)}{\Gamma(2s)} \exp\left(-\frac{1}{2}x^4\right) \sum_{n=0}^{\infty} \frac{x^{4n+4s-1}}{n! \Gamma(n+2s) \Gamma(J-n)}. \quad (16)$$

When J is a positive integer, this sum truncates and we find that $G(x)$ can be expressed as a linear combination of the squares of the J quasi-exact eigenfunctions of the Hamiltonian H in Eq. (3). For example, when $J=3$, we have

$$\begin{aligned} G(x) &= \frac{1}{\Gamma(2s)} \exp\left(-\frac{1}{2}x^4\right) x^{4s-1} \left[1 + \frac{x^4}{s} + \frac{x^8}{2s(2s+1)}\right] \\ &= \Gamma(2s) \left(\frac{2s+1}{4s+1} [\psi_0(x)]^2 + \frac{s}{4s+1} [\psi_+(x)]^2 + \frac{s}{4s+1} [\psi_-(x)]^2 \right), \end{aligned} \quad (17)$$

where $\psi_0(x)$ and $\psi_{\pm}(x)$ are taken from Eq. (11). We emphasize that this result is highly non-trivial. Expressing $G(x)$ as a linear combination of the squares of the eigenfunctions requires that one solve an overdetermined system of $2J-1$ equations for J expansion coefficients.

Let us now investigate the properties of the weight function $w(E)$. From the polynomials $P_n(E)$ we can calculate the moments of $w(E)$. Let a_n represent the $2n$ th moment of $w(E)$:

$$a_n = \int dE w(E) E^{2n}. \quad (18)$$

(Because the polynomials have parity symmetry we know that the odd moments vanish.) We are free to normalize $w(E)$ so that its zeroth moment is unity:

$$a_0 = 1. \quad (19)$$

The remaining moments can then be determined algebraically:

$$\begin{aligned} a_1 &= 32(J-1)s, \\ a_2 &= 32^2(J-1)s(3Js-5s+J-2), \\ a_3 &= 32^3(J-1)s(15J^2s^2-60Js^2+61s^2+15J^2s-67Js+74s+4J^2-19J+22), \\ a_4 &= 32^4(J-1)s(105J^3s^3-735J^2s^3+1743Js^3-1385s^3+210J^3s^2-1596J^2s^2+4038Js^2) \end{aligned} \quad (20)$$

$$\begin{aligned}
& -3372s^2 + 147J^3s - 1179J^2s + 3114Js - 2688s + 34J^3 - 282J^2 + 765J - 674), \\
a_5 = & 32^5(J-1)s(945J^4s^4 - 10080J^3s^4 + 40950J^2s^4 - 74400Js^4 + 50521s^4 + 3150J^4s^3 \\
& - 35910J^3s^3 + 153990J^2s^3 - 292154Js^3 + 205228s^3 + 4095J^4s^2 - 48960J^3s^2 + 218337J^2s^2 \\
& - 427524Js^2 + 307860s^2 + 2370J^4s - 29306J^3s + 134373J^2s - 269085Js + 197206s + 496J^4 \\
& - 6272J^3 + 29292J^2 - 59531J + 44134).
\end{aligned}$$

These moments have some interesting mathematical properties. For example, all the moments a_n , $n \geq 1$, have a factor of $(J-1)s$. Furthermore, in the residual factor the coefficient of $(Js)^{n-1}$ is $(2n-1)!!$ and the coefficient of s^{n-1} is the n th Euler number E_n .¹⁰

The outstanding property of the moments a_n concerns their rapid rate of growth. This rate of growth can be determined using the fact that there is a simple relationship between the moments a_n and the coefficients b_{n-1} of $P_{n-2}(E)$ in the recursion relation (6). Specifically, the Taylor series

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \quad (21)$$

whose coefficients are the moments in Eq. (18), is equivalent to a continued fraction

$$f(z) = 1/(1 - b_1 z / (1 - b_2 z / (1 - b_3 z / (\dots))))), \quad (22)$$

whose coefficients are b_n .¹¹ Since b_n is a cubic polynomial in n we deduce that the moments a_n grow like $(3n)!$.¹¹

It is unusual to find orthogonal polynomials whose weight functions have moments that grow so rapidly. The classical orthogonal polynomials, such as the Hermite polynomials, typically have moments that grow like $n!$. This is also true of discrete versions of the classical orthogonal polynomials, such as the Hahn polynomials.¹² The Euler and Bernoulli polynomials are distinctive¹³ in that their moments grow like $(2n)!$. However, the polynomials $P_n(E)$ associated with quasi-exact solvability are of an entirely new type due to the rapid rate of growth of their moments. Carleman's condition states that when the moments grow faster than $(2n)!$, the moment problem is not guaranteed to have a unique solution.¹⁵ Almost certainly, the weight function $w(E)$ is not unique! This nonuniqueness corresponds to a kind of gauge invariance that underlies these quasi-exactly solvable systems. Indeed one may conjecture that the nonuniqueness of the weight function is related to the Lie algebraic symmetry of quasi-exact solvability.

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⁷Here, *monic* means that the coefficient of the highest power of E is one.

⁸*Higher Transcendental Functions*, edited by A. Erdélyi (McGraw-Hill, New York, 1953), Vol. II, Chap. 10.

⁹Note that the actual domain of integration over E in (12) is from $-\infty$ to $+\infty$, but depending on the weight function this domain may be further truncated. Knowing the domain corresponds to knowing the explicit weight function, which is in general a difficult problem. However, knowledge of the domain is unnecessary for our algebraic approach.

¹⁰*Handbook of Mathematical Functions*, edited by M. Abramowitz and I. A. Stegun (National Bureau of Standards, Washington, 1970), Chap. 23.

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¹³The Euler and Bernoulli polynomials differ from the classical orthogonal polynomials in that they do not satisfy a

differential equation. This is likely to be true of the polynomials $P_n(E)$ discussed in this paper. Indeed, if the domain of the weight function for a set of orthogonal polynomials is the whole real line then the polynomials *must* be the Hermite polynomials (see Ref. 14).

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